

Richard Kim

San Diego, CA | 858-883-1416 | richard.kim1026@gmail.com | [LinkedIn](#) | US Citizen

Education

University of California, San Diego

Expected Dec. 2026

Bachelor of Science, Computer Engineering

- Relevant Coursework: Signals and Systems, Microelectronics, Digital Circuit Design, Computer Architecture, Algorithms, Data Structures, Systems Programming, Control Systems, Power Electronics

Experience

Software Engineer Intern

Y STEM and Chess – Boise, Idaho

July 2026 – Present

- Spearheaded an automated SOC 2 compliance platform by extracting and normalizing JSON schemas across AWS, GitHub, and Okta into a centralized, real-time summary for auditors.

Product Engineer | Postura (Smart Posture Monitor) | Engineering Capstone

UC San Diego – La Jolla, CA

Mar. 2026 – June 2026

- Developed C++ firmware for ESP32 microcontroller to acquire and filter spatial data from Time-of-Flight (ToF) sensor, implementing MQTT over Wi-Fi to publish real-time posture telemetry to a frontend web dashboard.
- Architected embedded hardware system, integrating the microcontroller, ToF sensor arrays, and power management components to ensure reliable data transmission and system stability.
- Designed and iterated mechanical enclosures using Fusion 360, optimizing the physical layout for seamless and comfortable integration of the electronics suite into a standard seat cushion.
- Conducted targeted user interviews with students and library staff to understand ergonomic habits and pain points, translating qualitative feedback into actionable hardware and software requirements.

Store Associate

The UPS Store – San Diego, CA

July 2024 – Oct. 2024

- Managed and streamlined customer service processes for returns, shipments, and store operations
- Carefully organized and assembled packages with high accuracy
- Maintained professionalism while resolving customer concerns effectively

Mechanical Team Member | Miramar Engineering Club

San Diego Miramar College – San Diego, CA

Oct. 2022 – May 2024

- Designed and modeled custom mounting brackets for hovercraft lift fans, ensuring secure integration with the main chassis under operational loads
- Developed CAD designs for custom rocket components, refining geometries based on physical constraints, material properties, and aerodynamic considerations
- Iteratively prototyped and optimized bracket designs, improving structural integrity, manufacturability, and ease of assembly
- Collaborated with engineering team to evaluate design performance and implement improvements across multiple design revisions

Projects

AI Thermal Detection System

[github](#)

- Developed a full-stack IoT dashboard using Python and FastAPI to capture and visualize live hardware sensor data streamed via MQTT from an ESP32 microcontroller.
- Engineered a custom authentication system backed by a relational MySQL database, implementing bcrypt for secure password hashing and UUIDs for persistent, cookie-based session management.
- Designed a containerized deployment pipeline using Docker and Docker Compose to reliably orchestrate the backend web server, database, and hardware communication channels.

SHA-256/Bitcoin Hash

Jun. 2025 - Sept. 2025

- Designed FPGA-based Bitcoin hash in SystemVerilog with 4-way parallel nonce processing to maximize hash throughput
- Achieved 635 clock cycles per hash for standard 80-byte block headers at maximum frequency of 70.88 MHz
- Implemented optimized SHA256 compression and message scheduling functions supporting variable-length inputs for flexible block header processing

Skills

Languages: C, C++, SystemVerilog, MATLAB, Python, Javascript, HTML, CSS, PyTorch, Docker, MySQL

Software: LTSpice, ModelSim, Intel Quartus Prime, Fusion360, Git, Linux/Unix, Cadence Virtuoso

Tools/Hardware: Oscilloscope, Multimeter, Function generator, Power supply, Soldering, ESP32, Arduino